

Garfield++ Gas Transport Parameter Scientific Plotting and Analysis Workbench

User Manual - English Edition

For `garfield_gas_workbench_pro_en.html` and `garfield_gas_workbench_pro.html`

2026-07-13

1. Document Overview

This manual applies to the latest single-file web application:

`garfield_gas_workbench_pro_en.html`

The same functions are also available in:

`garfield_gas_workbench_pro.html`

The application reads ASCII `.gas` files generated by Garfield, Garfield++, or Magboltz directly in a web browser. It supports multi-file comparison, physical-unit conversion, uncertainty visualization, interpolation, normalization, scans, fitting, two-dimensional plots, multi-panel figures, and publication-quality export.

No installation of Garfield++, ROOT, Python, or a web server is required. Gas files are read locally by the browser and are not automatically uploaded to the Internet.

1.1 Typical Applications

- Compare drift velocity, diffusion, Townsend coefficient, and attachment coefficient for different gas mixtures.
- Compare transport parameters at different temperatures, pressures, magnetic fields, or field grids.
- Study the effective Townsend coefficient $\alpha_{\text{eff}} = \alpha - \eta$ and locate its zero crossings.
- Scan a transport parameter as a function of gas fraction, pressure, or temperature.
- Produce figures and data files for papers, group meetings, presentations, and analysis reports.
- Save a complete analysis project and continue editing it later.

1.2 Recommended Browsers

Use a recent version of Google Chrome, Microsoft Edge, or Mozilla Firefox. A desktop browser with a screen width of at least 1280 px is recommended.

Because the application is a single HTML file, it can be opened by double-clicking it. If the browser restricts local storage, plotting templates may not be saved permanently. The **Save Full Project** function can still be used in that case.

2. Quick Start

2.1 Minimal Workflow

1. Open `garfield_gas_workbench_pro_en.html`.
2. Drag one or more `.gas` files into the upload area, or click **Add Files**.
3. In **Main Plot Settings - Data and Comparison**, select the X variable and the left Y parameter.
4. In the loaded-file list, enable the curves to display and adjust their order, color, line style, and marker style.
5. Use the mouse wheel, pan, or box zoom in the interactive plot.
6. In **Figure Size and Export**, choose a size preset and export PNG, SVG, or PDF.

2.2 Recommended Multi-Gas Comparison Workflow

Assume that several files correspond to increasing SF6 fraction:

1. Load all files at once.
2. Select **Specified Gas Fraction** as the automatic sort key.
3. Choose SF6 as the gas component and select ascending order.
4. Click **Sort**.
5. In the legend settings, enable **Gas Composition** and disable **File Name**.
6. Apply a colorblind-friendly palette and uniform line width and marker size.
7. Select E/N for the X axis and $\alpha_{\text{Eff}} = \alpha - \eta$ for the Y axis.
8. Enable logarithmic X or Y scaling when physically appropriate.
9. Export SVG or PNG using the single-column or double-column paper preset.

3. Supported Files and Parsing Rules

3.1 Supported File Type

The program is intended primarily for ASCII `.gas` files from Garfield, Garfield++, or Magboltz. A valid file should normally contain:

- Version
- GASOK bits
- Identifier
- Dimension
- E fields
- E-B angles
- B fields
- Mixture
- The gas tables follow:
- PGAS and TGAS in the footer

If the gas-table marker, footer, or grid description cannot be found, the application reports a parsing error.

3.2 Multidimensional Grids

The application can read gas tables containing:

- Electric-field grids
- Magnetic-field grids
- Electric-field-to-magnetic-field angle grids
- Excitation channels
- Ionization channels

The main plot displays one magnetic-field and one angle slice at a time. When multiple files have different grids, the nearest available magnetic-field and angle slice is selected independently for each file.

3.3 Pressure and Temperature

The footer is used to read:

- Pressure **PGAS**, treated internally in Torr
- Temperature **TGAS**, in kelvin

The gas number density is calculated using the ideal-gas relation:

$$N = \frac{p}{k_{\text{B}}T}.$$

Pressure is converted to pascals before this calculation.

4. Interface Layout

The interface is divided into five main areas:

1. **Files, Sorting, and Projects** - file management, order, visibility, curve styles, and project files.

2. **Main Plot Settings** - data, axes, titles, legend, uncertainties, and annotations.
3. **Interactive Main Plot** - curve display, zooming, panning, and value inspection.
4. **Figure Size and Export** - export size, DPI, background, and output format.
5. **Analysis and Batch Processing** - statistics, derived formulas, fitting, scans, heat maps, multi-panel figures, and raw-data inspection.

5. Files, Sorting, and Projects

5.1 Adding and Managing Files

Click **Add Files** to select multiple `.gas` files. Files added later are appended to the existing list rather than replacing it.

Each file entry supports:

- Show or hide its curve
- Set it as the current file
- Move it up or down
- Remove it
- Set a custom legend label
- Change its color
- Change line style, marker style, line width, marker size, and opacity
- Enable or disable marker fill

The current file determines the parameter list, magnetic-field and angle controls, raw-data view, and several analysis functions.

5.2 Manual Sorting

Use **Move Up** and **Move Down** for each file. File order affects:

- Curve drawing order
- Legend order
- File order in batch analyses

5.3 Automatic Sorting

Automatic sorting supports:

- Load order
- File name
- Pressure
- Temperature
- A specified gas fraction

When **Specified Gas Fraction** is selected, choose a component such as SF₆, CO₂, or i-C₄H₁₀. Sorting can be ascending or descending.

Composition values are taken primarily from the **.gas** Mixture array and are cross-checked against the Identifier for display and validation.

5.4 Color Palettes

Built-in palettes include:

- High-contrast colors
- Colorblind-friendly colors
- Black-and-white printing styles
- Sequential gradients

Apply Palette reassigns colors according to the current file order. It overwrites manually selected colors.

5.5 Quick Visibility Controls

Available actions include:

- Show all
- Hide all
- Show only the current file
- Invert selection

These controls are useful when many files are loaded.

5.6 Restore Default Styles

Restore Default Styles resets colors, line styles, marker styles, line widths, marker sizes, opacity, and marker fill. It does not remove files or change the selected physical quantities.

5.7 Saving and Loading a Full Project

Save Full Project exports a project file containing:

- Original **.gas** file text
- File order and visibility
- Per-curve styling
- Main-plot settings
- Custom derived formulas
- Annotations
- Other analysis state

Project files can be large because they include the original gas-file contents. After loading a project, the original **.gas** files do not need to be selected again.

5.8 Undo and Redo

The toolbar provides **Undo** and **Redo** for file order, styles, annotations, and selected settings.

Keyboard shortcuts:

- Ctrl+Z or Command+Z: Undo
- Ctrl+Shift+Z or Command+Shift+Z: Redo

5.9 File Integrity and Parsing Report

Expand **File Integrity and Parsing Report** to inspect:

- File-format version
- Grid dimensions
- Number of gas-table records
- Expected and parsed numeric-value counts
- Extra values
- GASOK flags
- Basic consistency between Identifier and Mixture
- Grid monotonicity and abnormal-value warnings

This report is the first place to check when a parameter is missing from the Y-axis list or a file behaves unexpectedly.

6. Main Plot Settings: Data and Comparison

6.1 X-Axis Variables

Three X-axis variables are supported:

- Actual electric field E [V/cm]
- Reduced electric field E/p [V/(cm*Torr)]
- Reduced electric field E/N [Td]

The conversions are:

$$E = (E/p)p,$$

$$E/N = \frac{E}{N}.$$

E/N is displayed in townsend, where $1 \text{ Td} = 10^{-21} \text{ V}\cdot\text{m}^2$.

6.2 Magnetic-Field and Angle Slices

- **E-B Angle Slice** selects the target field angle.
- **Magnetic Field B Slice** selects the target magnetic field.

A file with only one angle or magnetic field has only one option. If files use different grids, the application selects the nearest available slice. It does not automatically interpolate in magnetic field or angle.

6.3 Left and Right Y Axes

The left Y axis is the primary parameter. A second parameter may be assigned to the right Y axis to create a dual-axis plot.

Recommendations:

- Use dual axes only when the two quantities have a clear reason to be compared.
- Give both axes explicit names and units.
- Avoid arbitrary axis scaling that could create a misleading visual correlation.

6.4 Comparison Modes

The following modes are available.

Original Values

Display the data from each file without reference-based transformation.

Difference from Reference

$$\Delta y_i(x) = y_i(x) - y_{\text{ref}}(x).$$

Percentage Difference from Reference

$$\Delta y_i^{\%}(x) = \frac{y_i(x) - y_{\text{ref}}(x)}{y_{\text{ref}}(x)} \times 100\%.$$

If the reference value is zero, that point may be undefined.

Ratio to Reference

$$R_i(x) = \frac{y_i(x)}{y_{\text{ref}}(x)}.$$

Normalize Each Curve to Its Own Maximum

$$\tilde{y}_i(x) = \frac{y_i(x)}{\max_x y_i(x)}.$$

Normalize Each Curve at a Specified X Value

$$\tilde{y}_i(x) = \frac{y_i(x)}{y_i(x_0)}.$$

Enter x_0 in **Normalization X Value**. If x_0 is not an original grid point, the current interpolation method is used to obtain the normalization value.

6.5 Reference File

Difference, percentage, and ratio modes require a reference file. The reference file should cover the X range of the files being compared. Otherwise, some points cannot be calculated.

6.6 Common X Grid

Available choices are:

- **Original Grid for Each File** - preserve each curve's own grid.
- **Reference-File Grid** - map all curves to the reference file's X grid.
- **Union of All File Grids** - use the union of all X values.
- **Common Original Points Only** - keep only points present in every file.
- **Custom Uniform Grid** - enter start, end, and step values.

For ratio, difference, and consistent point-by-point comparison, the reference-file grid or a custom grid is usually recommended.

6.7 Interpolation Methods

Linear Interpolation

Interpolate linearly between neighboring data points.

Logarithmic-X Interpolation

Interpolate in $\log_{10}(x)$ space. This is useful when the positive X values span several orders of magnitude. All X values must be positive.

No Interpolation - Common Original Points Only

Use only original points with matching X values. This option is appropriate for strict quantitative comparisons where interpolation must be avoided.

Tooltips identify interpolated points. When using interpolation in a paper or formal analysis, document the chosen method.

6.8 Curve Connection Modes

Supported modes:

- Straight-line connection
- Original points only
- Step plot
- Smoothed visual curve

The smoothed curve is for presentation only. It is not a new Garfield++ calculation. Statistics, exported data, and quantitative conclusions should preferentially use original points or explicitly generated interpolation points.

6.9 File Uncertainties

If the `.gas` format contains uncertainty information, the program can display:

- Error bars
- Error bands
- Error bars and bands together

Uncertainties are shown only in **Original Values** mode. Some parameters and some gas-table versions do not contain uncertainty information.

6.10 Point Labels and Crosshair Readout

- **Show Point Labels** displays selected numeric labels rather than labeling every point when curves are dense.
- **Crosshair and Synchronized Readout** displays values from multiple curves near the cursor position.

7. Main Plot Settings: Axes and Titles

7.1 Linear and Logarithmic Scales

The X axis, left Y axis, and right Y axis can independently use linear or logarithmic scaling.

Logarithmic axes display only positive values. Zero and negative values are filtered. Therefore:

- `alphaEff` can cross zero and is generally unsuitable for a logarithmic Y axis.
- Differences and percentage changes may be negative and are unsuitable for a logarithmic Y axis.
- With a logarithmic X axis, `E`, `E/p`, or `E/N` must be positive.

7.2 Manual Axis Ranges

After enabling **Manual Axis Range**, enter:

- X minimum and maximum
- Left-Y minimum and maximum
- Right-Y minimum and maximum

A practical workflow is to zoom and pan first, then click **Copy Current View to Range Fields**. This makes it easy to use exactly the same limits in several figures.

Auto Range disables manual limits and restores data-based limits.

7.3 Axis Reversal and Plot Frame

Available options include:

- Reverse X axis
- Reverse left Y axis
- Reverse right Y axis
- Show top axis
- Show right frame

Gas-transport curves normally do not require reversed axes unless a specific presentation convention is needed.

7.4 Tick-Label Formatting

Supported formats:

- Automatic
- Scientific notation
- Fixed decimals
- 10^n notation

The number of decimal places can be set in fixed-decimal mode.

7.5 Grid and Tick Density

Major and minor grid lines can be enabled separately, and the number of major and minor ticks can be adjusted. For publication figures:

- Use light major grid lines.
- Limit minor-grid density.
- Avoid overcrowded tick labels.

7.6 Titles and Axis Labels

The following text can be customized:

- Main title
- Subtitle
- X-axis label
- Left-Y-axis label
- Right-Y-axis label
- Title font size
- Axis-title font size
- Tick font size

If a field is left blank, the current parameter name and unit are used automatically.

8. Main Plot Settings: Legend and Styles

8.1 Legend Content

The legend can combine:

- File name
- Gas composition
- Temperature
- Pressure

Each curve can also use a custom legend label. This is useful for shortening long file names, for example:

```
SF6 = 0%
SF6 = 1%
SF6 = 3%
SF6 = 5%
```

8.2 Legend Position

Supported positions:

- Bottom
- Top
- Left
- Right
- Inside, upper-right corner

Additional controls include:

- Number of legend columns
- Legend font size
- Legend title
- Background opacity
- Border visibility
- Reverse legend order

An in-plot legend may cover data, so inspect the selected position carefully.

8.3 Per-Curve Styles

Each file can independently define:

- Color
- Line style: solid, dashed, dotted, or dash-dot
- Marker: circle, square, diamond, triangle, cross, plus, or none
- Line width
- Marker size
- Opacity
- Marker fill

For black-and-white printing, combine different line styles and markers rather than relying on grayscale alone.

8.4 Uniform Styling

Enter a common line width, marker size, and opacity, then click **Apply Uniform Size** to apply them to all files.

9. Auxiliary Lines and Annotations

The application can add:

- Vertical lines
- Horizontal lines
- Shaded X intervals
- Shaded Y intervals
- Text labels
- Arrows

Required values depend on the annotation type:

- Vertical line: X1
- Horizontal line: Y1
- X shading: X1 and X2
- Y shading: Y1 and Y2
- Text: position and text content
- Arrow: start and end coordinates

Annotation color can be selected. Added annotations appear in a list and can be removed individually. They are saved with the project and included in figure exports.

10. Interactive Main Plot

10.1 Mouse Operations

- Mouse wheel: zoom around the cursor position
- Pan mode: drag with the left mouse button
- Box-zoom mode: draw a rectangle around the desired region
- Double-click: restore the full view
- **Reset** button: restore automatic limits

10.2 Keyboard Shortcuts

- R: reset the plot when the cursor is not inside a text input
- **Ctrl+S** or **Command+S**: export PNG
- **Ctrl+Z** or **Command+Z**: undo
- **Ctrl+Shift+Z** or **Command+Shift+Z**: redo

The browser may override some shortcuts. If **Ctrl+S** opens the browser's Save Page dialog, use the page's **Export PNG** button instead.

11. Supported Physical Quantities and Conversions

11.1 Drift Velocity

- v_E [cm/ns] = raw.ve * 10^{-3}
- v_E [cm/us] = raw.ve
- v_{B_perp} [cm/ns] = raw.vb * 10^{-3}
- v_{ExB} [cm/ns] = raw.vexb * 10^{-3}

11.2 Diffusion

Let pressure p be expressed in Torr:

$$D_L = \frac{D_{L,raw}}{\sqrt{p}}, \quad D_T = \frac{D_{T,raw}}{\sqrt{p}}.$$

The application provides:

- DL [sqrt(cm)] and DT [sqrt(cm)]
- DL [um/sqrt(cm)] and DT [um/sqrt(cm)]

11.3 Townsend and Attachment Coefficients

The stored values are restored from logarithmic form:

$$\frac{\alpha}{p} = \exp(\text{raw.alpha}),$$

$$\frac{\eta}{p} = \exp(\text{raw.eta}).$$

Therefore:

$$\alpha = p \exp(\text{raw.alpha}),$$

$$\eta = p \exp(\text{raw.eta}).$$

The effective Townsend coefficient is calculated by the web application:

$$\alpha_{\text{eff}} = \alpha - \eta.$$

The corresponding reduced coefficient is:

$$\frac{\alpha_{\text{eff}}}{p} = \frac{\alpha}{p} - \frac{\eta}{p}.$$

11.4 Number-Density-Reduced Quantities

The application provides:

$$\frac{\alpha}{N}, \quad \frac{\eta}{N}, \quad \frac{\alpha - \eta}{N}.$$

Number density is converted from m^{-3} to cm^{-3} for these calculations.

11.5 Other Derived Quantities

- Electron mobility:

$$\mu_e = \frac{v_E}{E}.$$

- DL/vE
- DT/vE
- Ion mobility
- Lorentz angle
- Ion dissociation coefficient
- Six diffusion-tensor components
- Total excitation and ionization rates
- Individual excitation- and ionization-channel rates

The Lorentz angle and some rate quantities retain the units used in the source file. Confirm the source and units in **Raw Data and Parameter Sources** before quantitative use.

12. Analysis: Feature Points and Statistics

12.1 Statistical Interval

An X minimum and maximum can be entered for the statistical interval. Leave both fields blank to use the full curve range.

12.2 Statistical Summary

For each visible curve, the program calculates:

- Number of points
- Minimum value and its X coordinate
- Maximum value and its X coordinate
- Mean
- Median
- Trapezoidal integral
- Zero crossings or sign changes

The numerical integral is:

$$\int y \, dx \approx \sum_i \frac{y_i + y_{i+1}}{2} (x_{i+1} - x_i).$$

12.3 Feature Points

The program identifies:

- Minimum
- Maximum
- Zero crossings or sign changes
- An approximate plateau start for drift-velocity curves

A zero crossing is estimated by linear interpolation between neighboring points with opposite signs. Plateau detection is a screening heuristic and should not replace a dedicated physical criterion.

13. Analysis: Derived Formulas

13.1 Available Variables

Custom expressions may use:

E
EoverP
EoverN
alpha
eta
alphaEff
alphaReduced
etaReduced
veCmNs
veCmUs
DL
DT
pressure
temperature
N
Math

Math provides JavaScript mathematical functions, for example:

```
Math.log10(alphaEff)
Math.sqrt(DT * DT + DL * DL)
```

13.2 Examples

Create an effective coefficient normalized by number density:

Parameter name: `alphaEff/N`
Expression: `alphaEff/(N/1e6)`

Create an inverse-drift-velocity scale:

Parameter name: `1/vE`
Expression: `1/veCmNs`

13.3 Safety Restrictions

Expressions are evaluated locally. The application blocks semicolons, code blocks, arrow functions, and objects such as `window`, `document`, `fetch`, `eval`, `Function`, and `localStorage`. Even so, use only mathematical expressions that you understand and trust.

14. Analysis: Curve Fitting

14.1 Fit Settings

Select:

- File
- Parameter

- Fit model
- Polynomial order
- X minimum and maximum

14.2 Polynomial Fit

The model is:

$$y = \sum_{k=0}^n c_k x^k.$$

Coefficients are solved using normal equations. High polynomial orders or very large X ranges may be numerically unstable. Avoid high-order fits without a clear need.

14.3 Exponential Fit

The model is:

$$y = A \exp(Bx).$$

Only points with $y > 0$ are used. Parameters are obtained from a linear fit to $\ln y$.

14.4 Fit Results

The application reports:

- Fit expression
- Coefficient of determination R^2
- Number of points used

The fitted curve is overlaid on the main plot as a separate dashed line. It is a web-calculated result, not original Garfield++ data.

15. Analysis: Pressure, Temperature, and Composition Scans

This function uses a file property as the X axis instead of drawing one curve per file.

15.1 Scan X Variable

Available scan variables:

- Pressure

- Temperature
- A specified gas fraction

15.2 Fixed Field Variable

The fixed field can be:

- E
- E/p
- E/N

After a fixed value is entered, the program interpolates the selected parameter in each file and plots it as a function of pressure, temperature, or gas fraction.

15.3 Recommendations

Before performing a scan, verify that:

- The physical parameter definitions are consistent across files.
- The fixed field value lies inside the valid range of every file.
- The interpolation method is appropriate.
- File order and composition identification are correct.

16. Analysis: Heat Maps, Contours, and 3D Projection

16.1 Applicability

The current file should contain multiple magnetic-field points or multiple E-B angle points. Otherwise, the 2D data reduce to a single row or column.

16.2 Plane Selection

Available planes:

- E versus B
- E versus E-B angle

Use **Fixed Remaining-Dimension Index** to select the third dimension that is not being plotted.

16.3 Display Modes

- Two-dimensional heat map
- Contour plot
- Three-dimensional surface projection

The 3D display is an SVG-based isometric projection for trend inspection. It is not an interactive WebGL model.

17. Analysis: Multi-Panel Figures and Batch Export

17.1 Multi-Panel Layouts

Supported layouts:

- 1 x 2
- 2 x 2
- 2 x 3
- 3 x 3

After several parameters are selected, each parameter is drawn in its own panel using the same file colors.

17.2 Batch Export

Export Selected Parameters as SVG ZIP generates one SVG per selected parameter and packages them into a ZIP archive. This is useful for producing a standard set of plots such as:

- Drift velocity
- Longitudinal diffusion
- Transverse diffusion
- Townsend coefficient
- Attachment coefficient
- Effective Townsend coefficient

18. Raw Data and Parameter Sources

The **Raw Data and Sources** tab is intended for checking parsing and unit conversion.

It can display:

- Current file
- A specific grid record
- Raw record values
- Converted physical quantities
- Parameter sources and conversion formulas
- Original file header

When a value appears abnormal, check in this order:

1. File pressure and temperature
2. Current magnetic-field and angle slices
3. Raw record
4. GASOK flags
5. Parameter conversion formula

6. Whether comparison, normalization, or interpolation is enabled

19. Figure and Data Export

19.1 Export Formats

Supported outputs include:

- PNG
- SVG
- PDF
- Browser print or Save as PDF
- Current-data CSV
- HTML analysis report
- Multi-parameter SVG ZIP

19.2 Size Presets

Built-in presets include:

- Custom
- Single-column paper figure: 8.5 x 6.2 cm
- Double-column paper figure: 17.8 x 11 cm
- 16:9 presentation
- 4:3 presentation
- Square

19.3 Size Units and DPI

Dimensions may be entered in:

- Pixels
- Centimeters
- Inches

Centimeters and inches are converted to pixels using:

$$N_{\text{px}} = L_{\text{in}} \times \text{DPI}.$$

For raster figures in papers, 300-600 DPI is common. SVG is resolution-independent and is generally preferred for papers and later editing.

19.4 Background

The background can be white or transparent. Transparent output is useful in presentations, but confirm that a journal submission system handles transparent PNG or SVG correctly.

19.5 Exported Content

Available choices:

- Plot, legend, and notes
- Plot only

Plot Only is useful when legends and annotations will be added later in layout software.

19.6 CSV Content

The current-data CSV includes:

- File name
- Identifier
- Pressure
- Temperature
- X variable and value
- Parameter name and value
- Uncertainty
- Interpolated-point flag

The CSV reflects the current main-plot processing state, including comparison mode, common grid, and interpolation settings.

19.7 HTML Analysis Report

The report contains:

- File information
- Current main plot
- Feature points
- Statistical summary
- Method notes

Open the report in a browser and print it to PDF when needed.

20. Plotting Templates versus Full Projects

20.1 Plotting Templates

A template mainly stores plotting settings such as:

- Axes
- Titles
- Legend
- Font sizes
- Comparison and interpolation settings

Templates are stored in browser `localStorage` and apply only to the current browser profile. Clearing browser data or using private browsing may remove them.

20.2 Full Projects

A full project stores the original gas files and all application state. It can be copied to another computer and reopened there. For long-term research, saving the full project is strongly recommended.

21. Recommended Research Workflows

21.1 Publication-Quality Multi-Gas Comparison

1. Load all files.
2. Inspect the parsing report.
3. Automatically sort by gas fraction.
4. Apply a colorblind-friendly palette.
5. Use a common line width of 2-3 and marker size of 4-6.
6. Show only composition and pressure in the legend.
7. Apply identical axis ranges across related figures.
8. Disable unnecessary minor grids.
9. Select a single-column or double-column preset.
10. Export SVG as the primary figure and PNG as a preview.

21.2 Studying the Effective Townsend Zero Crossing

1. Select `alphaEff = alpha - eta`.
2. Use a linear Y axis.
3. Inspect zero crossings in **Feature Points and Statistics**.
4. Add a horizontal guide at `Y = 0`.
5. Zoom into the crossing region.
6. Export CSV and verify the original neighboring values.

21.3 Comparing Different Pressures

To compare transport properties at the same reduced field:

1. Select `E/N` or `E/p` as the X axis.
2. Use the reference-file grid.
3. Select a suitable interpolation method.
4. Use ratio or percentage mode.
5. Choose the baseline-pressure file as the reference.
6. Show pressure and temperature in the legend.

21.4 Finding a Useful Gas-Fraction Range

1. Load files with different mixture fractions.
2. Sort automatically by the target gas fraction.
3. Open the composition-scan function.
4. Fix E/N.
5. Scan `alphaEff`, drift velocity, and diffusion separately.
6. Add a shaded acceptable operating interval.
7. Export the scan plot and current-data CSV.

22. Troubleshooting

22.1 File Cannot Be Read

Possible causes:

- The file is not an ASCII `.gas` file.
- The `gas tables follow:` is missing.
- The `H Extr:` footer is missing.
- `Dimension` does not match the number of data values.
- The file is truncated or damaged.

Read the status message and parsing report, then inspect the file header and footer in a text editor.

22.2 A Parameter Is Missing from the List

Possible causes:

- The GASOK flag marks the parameter as unavailable.
- The file has no corresponding channel.
- Channel definitions differ between the current file and other files.

Switch the current file and inspect the parameter list again.

22.3 No Data Are Visible

Check whether:

- All files are hidden.
- A logarithmic axis filtered zero or negative values.
- Manual minimum and maximum values were entered in reverse order.
- The selected magnetic-field or angle slice contains data.
- The reference file covers the required range in comparison mode.
- A custom grid lies outside the data range.
- No common points exist in no-interpolation mode.

22.4 Ratio or Percentage Values Are Unstable

If the reference value is close to zero, ratios and percentage differences can become extremely large or undefined. Inspect the reference curve and restrict the analysis interval.

22.5 Error Bars Are Not Shown

- The file may not contain uncertainty information.
- The selected parameter may not have implemented uncertainty conversion.
- The comparison mode may not be **Original Values**.
- Uncertainty values may be zero or non-finite.

22.6 Legend Composition Is Unexpected

Composition display combines Identifier and the Mixture array. Different Magboltz versions or custom files may use different naming conventions. Verify the original file header and Mixture values.

22.7 Templates Cannot Be Saved

The browser may block local storage. Possible solutions:

- Exit private-browsing mode.
- Allow local storage for the page.
- Use **Save Full Project** instead.

22.8 PDF Text or Layout Is Unsatisfactory

Try one of the following:

- Export SVG and convert it to PDF with Inkscape, Illustrator, or Power-Point.
- Use the browser's Print or Save as PDF function.
- Increase export size and font sizes.
- Avoid placing a very long legend inside the plot.

23. Limitations and Scientific Cautions

1. The application parses, converts, and visualizes data; it does not replace Garfield++ or Magboltz calculations.
2. Interpolation, fitting, smoothing, and feature-point detection are secondary processing results and must be distinguished from original data.
3. Magnetic-field and angle slices use nearest-neighbor selection, with no automatic interpolation in those dimensions.

4. Units for Lorentz angle, excitation rates, ionization rates, and some tensor quantities depend on the source-file definition. Verify them before quantitative use.
5. Number-density conversion assumes that pressure and temperature meta-data are correct and that the ideal-gas approximation is appropriate.
6. High-order polynomial fits can be numerically unstable. A high R^2 does not imply physical validity.
7. Dual-Y-axis plots can be visually misleading. Label both axes and units clearly.
8. Custom expressions should contain only trusted mathematical formulas.
9. A project file may contain complete original gas data. Check for sensitive or unpublished information before sharing it.
10. For reproducibility, archive the original `.gas` files, project file, exported CSV, and plotting settings before publication.

24. Parameter Quick Reference

Category	Parameter	Source or conversion in the application
Drift velocity	<code>vE [cm/ns]</code>	<code>raw.ve * 10⁻³</code>
Drift velocity	<code>vE [cm/us]</code>	<code>raw.ve</code>
Drift velocity	<code>vB_perp</code>	<code>raw.vb * 10⁻³</code>
Drift velocity	<code>vExB</code>	<code>raw.vexb * 10⁻³</code>
Diffusion	<code>DL [sqrt(cm)]</code>	<code>raw.dl/sqrt(p)</code>
Diffusion	<code>DT [sqrt(cm)]</code>	<code>raw.dt/sqrt(p)</code>
Diffusion	<code>DL, DT [um/sqrt(cm)]</code>	Previous result multiplied by 10^4
Multiplication	<code>alpha</code>	<code>p*exp(raw.alpha)</code>
Multiplication	<code>alpha0</code>	<code>p*exp(raw.alpha0)</code>
Attachment	<code>eta</code>	<code>p*exp(raw.eta)</code>
Derived	<code>alphaEff</code>	<code>alpha - eta</code>
Reduced	<code>alpha/p</code>	<code>exp(raw.alpha)</code>
Reduced	<code>eta/p</code>	<code>exp(raw.eta)</code>
Reduced	<code>(alpha-eta)/p</code>	<code>alpha/p - eta/p</code>
Number-density reduced	<code>alpha/N, eta/N, alphaEff/N</code>	Use N converted to cm^{-3}
Derived	<code>mu_e</code>	<code>vE/E</code>
Derived	<code>DL/vE, DT/vE</code>	Calculated by the application
Other	Ion mobility	<code>raw.mu * 10⁶</code>
Other	Ion dissociation	<code>p*exp(raw.dissociation)</code>
Other	Diffusion tensor	<code>raw.tensor[i]/p</code>
Rates	Total excitation and ionization	Sum over channels

25. Recommended Data-Archive Structure

Create a separate directory for each study:

```
GasStudy_2026-07-13/  
|-- gas_files/  
|   |-- mixture_01.gas  
|   |-- mixture_02.gas  
|   |-- mixture_03.gas  
|-- project/  
|   |-- analysis_project.json  
|-- figures/  
|   |-- alpha_eff.svg  
|   |-- alpha_eff.png  
|   |-- drift_velocity.svg  
|-- data/  
|   |-- current_plot_data.csv  
|-- notes/  
    |-- conditions.md
```

Record at least:

- Garfield++ and Magboltz versions
- Gas fractions
- Pressure and temperature
- Penning settings
- Electric-field, magnetic-field, and angle grids
- Web-project file version
- Interpolation and comparison methods

26. Applicable Version

This manual was prepared for:

```
garfield_gas_workbench_pro_en.html  
garfield_gas_workbench_pro.html
```

English manual date: 2026-07-13.

If the web application is modified later, update the control names, conversion formulas, and listed limitations in this manual accordingly.